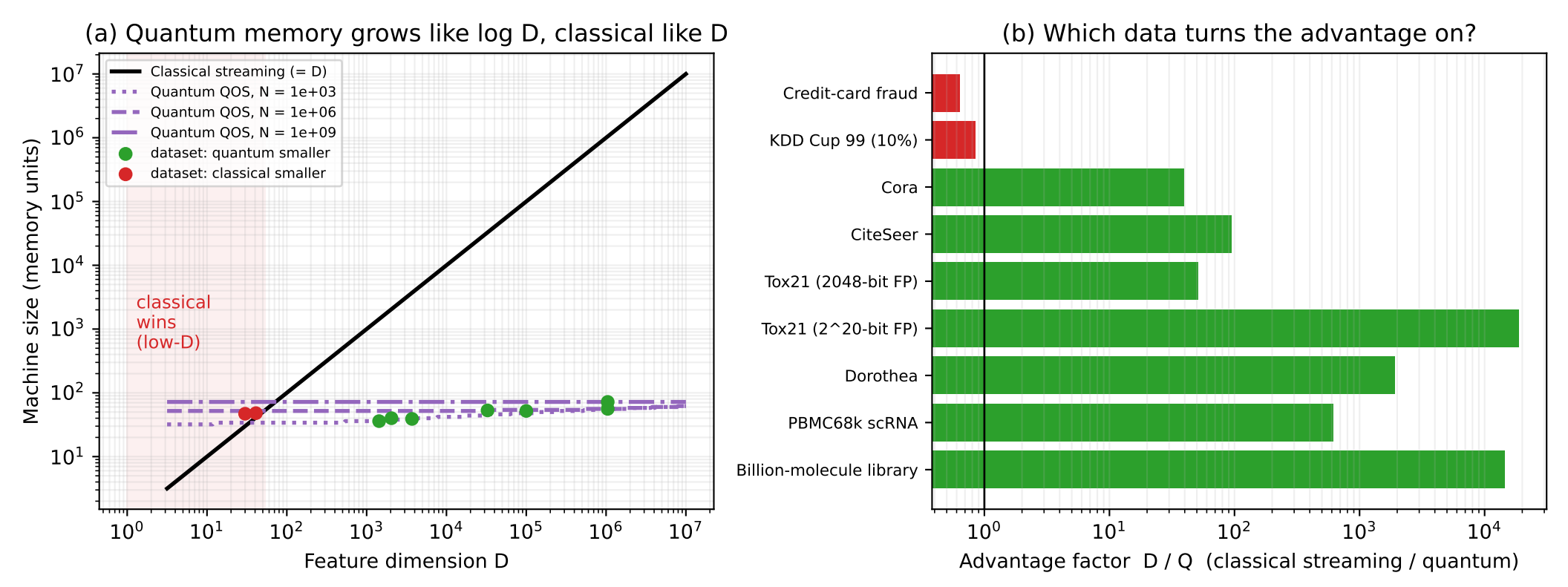
(a) Quantum memory grows like log D, classical like D
Classical streaming (= D)
Quantum QOS, N = 1e+03
Quantum QOS, N = 1e+06
Quantum QOS, N = 1e+09
dataset: quantum smaller
dataset: classical smaller
classical
wins
(low-D)
Machine size (memory units)
Feature dimension D
(b) Which data turns the advantage on?
Credit-card fraud
KDD Cup 99 (10%)
Cora
CiteSeer
Tox21 (2048-bit FP)
Tox21 (2^20-bit FP)
Dorothea
PBMC68k scRNA
Billion-molecule library
Advantage factor D / Q (classical streaming / quantum)